

Tessellations of meromorphic functions and the Riemann's existence theorem

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Abstract

A meromorphic function R , of degree $n \geq 2$, on a compact Riemann surface M provided with a cyclic order on its q distinct critical values, determines a homogeneous tessellation of the Riemann surface M , whose $2n$ tiles are topological q -gons with alternating colors. The tessellation provides a simple and straightforward visual description of the meromorphic function R . Conversely, assume a possibly non homogeneous tessellation $\mathcal{T}$ of a compact oriented C^1 surface $\mathcal{M}$, with tiles of alternating colors and a suitable labelling in the vertices of its tiles. Non homogeneous means that the tiles of $\mathcal{T}$ are ρ -gons, for different values of ρ . Then there exists a Riemann surface structure M on $\mathcal{M}$, a meromorphic function R and a cyclic order of its critical values, such that the tessellation of R on M topologically coincides with the original $\mathcal{T}$.

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MSC2020 Classification: 30C10 , 30A99 , 30C25 , 30F10

1 Introduction

The field of meromorphic functions on a compact Riemann surface M encodes the algebraical, analytical and geometrical, aspects of the complex structure of M . Our starting point is the following classical idea, that we call, the Schwarz–Klein’s algorithm:

*A meromorphic function $R : M \longrightarrow \widehat{\mathbb{C}}$, of degree $n \geq 2$,
provided with a cyclic order $\mathcal{L}_\gamma$ of its critical values,
determines a tessellation $\mathcal{T}_\gamma(R)$ of the Riemann surface M
whose $2n$ tiles are topological q -gons with alternating colors.*

Here $q \geq 2$ is the number of distinct critical values of R . Certainly, the algorithm requires a little bit of information; a *cyclic order* $\mathcal{L}_\gamma$ of the q critical values of R . This is provided by an oriented Jordan path γ in $\widehat{\mathbb{C}}$ running through the critical values of R . Hence, the input data of the algorithm is a triad $(M, R, \mathcal{L}_\gamma)$. The output is the homogeneous tessellation $\mathcal{T}_\gamma(R)$, it allows for a simple/straightforward visual description of the meromorphic function R . A homogeneous tessellation is a collection of copies of two topological q -gons¹, T and T' , that fit together to cover the surface M , without overlaps or gaps. Non homogeneous means that the tiles are ρ -gons, for different values of ρ .

As far as we known this idea has its roots in the works of H. A. Schwarz and F. Klein.

In search of an inverse result, let us consider a possibly non homogeneous tessellation $\mathcal{T}$ on a compact oriented C^1 surface $\mathcal{M}$, whose $2n \geq 4$ tiles, are topological ρ -gons with alternating colors.

How can we recognize that $\mathcal{T}$ is equivalent to a tessellation $\mathcal{T}_\gamma(R)$, up to orientation preserving homeomorphism, as in the Schwarz–Klein’s algorithm?

Around 2010, W. P. Thurston conducted a group discussion on what he called *the shape of rational maps*. As a result of the group discussion, S. Koch *et al.* [18] considering tessellations of $\mathbb{S}^2$, found specific conditions, namely that the underlying graph is globally and locally balanced, under which the tessellation can be recognized topologically as a generic branched covering.

In fact, Thurston’s notion of consistent labelling (a consequence of the global and local balance of the tessellation) and our notion of *consistent q -labelling* $\mathcal{L}_c$, provide a labelling of the vertices of the tessellation, consistent with the labelling of the vertices of the individual tiles as q -gons, also taking into account the orientation according to their coloring and another technical condition; see Definition 4 for full details.

In order to provide an answer to the previous question, let $M = (\mathcal{M}, J)$ be a compact Riemann surface, $\mathcal{M}$ denotes a compact oriented C^1 surface of genus g and $J : T\mathcal{M} \longrightarrow T\mathcal{M}$ is the integrable almost complex structure, $J \circ J = -Id$. The orientation of $\mathcal{M}$ coincides with the one induced by J . The main result is as follows.

Theorem 1 *Let $M = (\mathcal{M}, J)$ be a compact Riemann surface of genus g , and $n \geq 2$ be an integer number.*

¹The tiles T and T' arise as the complement of γ in $\widehat{\mathbb{C}}$: that is $\widehat{\mathbb{C}} \setminus \gamma = T \cup T'$.

- 1) A meromorphic function $R: M \rightarrow \widehat{\mathbb{C}}$, of degree n , provided with a cyclic order $\mathcal{L}_\gamma$ of its q critical values, determines a homogeneous tessellation

$$\mathcal{T}_\gamma(R) = \underbrace{T_1 \cup \dots \cup T_n}_{\text{blue tiles}} \cup \underbrace{T'_1 \cup \dots \cup T'_n}_{\text{gray tiles}} \subset M,$$

whose tiles are topological q -gons with alternating colors, and a consistent q -labelling $R^*\mathcal{L}_\gamma$.

- 2) Let $\mathcal{T}$ be a possibly non homogeneous tessellation of $\mathcal{M}$, whose $2n$ tiles are topological ρ -gons with alternating colors. Assume in addition that $\mathcal{T}$ is provided with a consistent q -labelling $\mathcal{L}_c$, such that $2 + 2g \leq \rho \leq q \leq 2n + 2g - 2$. Then $(\mathcal{T}, \mathcal{L}_c)$ determine a (non unique) Riemann surface M , a meromorphic function

$$R: M \rightarrow \widehat{\mathbb{C}},$$

and a Jordan path γ satisfying that the tessellation $\mathcal{T}_\gamma(R)$ is $\mathcal{T}$, up to orientation preserving homeomorphism of $\mathcal{M}$.

A cyclic order $\mathcal{L}_\gamma$ determines a consistent q -labelling $R^*\mathcal{L}_\gamma$ for the given meromorphic function R . Conversely, there are tessellations $\mathcal{T}$ that admit consistent q -labellings for different values of q ; see Example 2, Figures 1.c and 2.

The pair $(\mathcal{T}, \mathcal{L}_c)$ in assertion (2) gives rise to an analytical triple $(M, R, \mathcal{L}_\gamma)$, which in turn determines the pair $(\mathcal{T}_\gamma(R), R^*\mathcal{L}_\gamma)$, which is topologically orientation preserving equivalent to the original pair $(\mathcal{T}, \mathcal{L}_c)$.

However, by starting with an analytical triple $(M, R, \mathcal{L}_\gamma)$, it determines a pair $(\mathcal{T}_\gamma(R), R^*\mathcal{L}_\gamma)$, which in turn gives rise to an analytical triple $(\widetilde{R}, \widetilde{M}, \widetilde{\gamma})$ which generically is not analytically equivalent to the original (R, M, γ) .

In order to geometrically perform the gluing, implicit in assertion (2) of Theorem 1, we furnish the sphere with a holomorphic vector field $(\widehat{\mathbb{C}}, \partial/\partial w)$, that is with a translation structure, see Corollary 2. Moreover, by showing that the critical values are parameters involved in the construction, one can obtain many different triads $(M, R, \mathcal{L}_\gamma)$ from the same tessellation $(\mathcal{T}, \mathcal{L}_c)$.

A very brief list of technical and historical² comments is below.

- We attribute assertion (1) of Theorem 1 to H. A. Schwarz (recall his study about uniformization of polygonal regions, and the resulting tessellations [29], a contemporary description is in [35, ch. 5]) and to F. Klein following [16], [7]. Hence, we introduce the Schwarz–Klein’s algorithm in §2.1. A. Speiser [30] and R. Nevanlinna [27], considered tessellations for transcendental functions, also see R. Peretz [28].
- The case of $q = 3$ critical values is essentially a first step in the Theory of dessins d’enfants, due to G. V. Belyĭ [5] and A. Grothendieck [13], currently a highly developed subject. See the books S. K. Lando *et al.* [19], A. Degtyarev [8] and G. A. Jones *et al.* [15] remarkable for its historic review.
- Assertion (1) of Theorem 1 is a statement, using tessellations, of the Riemann’s existence theorem, compare with the versions in I. Bauer *et al.* [4], H. Volklein [34, Ch. 6], S. I. Lando *et al.* [19, p. 74] and R. Cavalieri *et al.* [6, p. 85].

²As usual, advances in the theory have been discovered independently by diverse authors.

- The role of the Jordan path γ in assertion (1) is to provide a cyclic order $\mathcal{L}_\gamma$ to the critical values of R ; see Corollary 3. On $M = \widehat{\mathbb{C}}$, the change of the order was studied by H. Habsch [14], see also S. Lando *et al.* [19, 5.4] for the polynomial case.
- The proof of assertion (2) can be described succinctly as “pullback the complex structure”, compare with S. Stoilow [31]. However, this road requires heavy analytic machinery. Our proof uses elementary surgery techniques, as Corollary 2 emphasizes. In fact, it can be widely applied to complex analytic functions, quadratic differentials K. Strebel [32, § 12.3] (we learned this surgery technique from him), H. G. Dias–Marin [9], and vector fields J. Muciño–Raymundo [25].
- In the case of $M = \widehat{\mathbb{C}}$, the meromorphic functions sharing a set of critical values was described algebraically by J. Mycielski [26], compare with the classical result of L. R. Goldberg [10]. Recently, K. Lazebnik [20] provides an analytic study of the paths $R^{-1}(\gamma)$.
- Meromorphic functions with q -critical values give rise to subfamilies of $F(w_1, \dots, w_q)$, the Riemann surfaces whose branch points lie over a finite set of singular values $\{w_1, \dots, w_q\} \subset \widehat{\mathbb{C}}$, where finite and/or infinite branchings are allowed; see R. Nevanlinna [27, ch. XI, §2]. In order to describe a meromorphic function $w(z)$, whose Riemann surface belongs to $F(w_1, \dots, w_q)$, the usual tool is its *Speiser graph*. See for instance A. A. Goldberg *et al.* [12, ch. 7 §4]. In the topological sense, a Speiser graph is the dual of a tessellation associated to $w(z)$.
- Tessellations have been applied to the study of differential equations, see D. Masoero [24], G. León–Gil *et al.* [21], A. Alvarez–Parrilla *et al.* [2], and references therein.

The article is organized as follows. In Section 2, we recall the definition of tessellation and its associated graphs. The Schwarz–Klein’s algorithm is introduced in §2.1. Section 3 contains the proof of assertion (2) of Theorem 1. In §3.1, we review the relationship between analytic and combinatorial structures involved in the work. A construction of meromorphic functions by surgery is provided in Corollary 2. Section 4 presents examples and a brief comment on future directions.

2 Tessellations and graphs on Riemann surfaces

Tessellations and graphs appear in many instances in the study of complex analytic functions and Riemann surfaces, with very intricate meanings and notations. We provide accurate ad hoc concepts.

Let $M = (\mathcal{M}, J)$ be a compact Riemann surface, as in the Introduction. Definitions 1–4 below are considered over the C^1 surface $\mathcal{M}$, hence they apply on M .

Definition 1 A *tessellation* of a surface $\mathcal{M}$ is a collection with alternating colors

$$\mathcal{T} = \underbrace{T_1 \cup \dots \cup T_n}_{\text{blue tiles}} \cup \underbrace{T'_1 \cup \dots \cup T'_n}_{\text{gray tiles}} \subset \mathcal{M}, \quad n \geq 2, \quad (1)$$

where the $2n$ tiles $\{T_\alpha, T'_\alpha\}_{\alpha=1}^n$ are open Jordan domains, such that:

- The union of their closures $\cup_{\alpha=1}^n (\overline{T_\alpha} \cup \overline{T'_\alpha})$ is $\mathcal{M}$.
- If the intersection of the closures of any two tiles is non-empty, then it consists of a finite number of simple paths (edges) and their extreme points called vertices.

iii) There are at least two vertices.

A tessellation $\mathcal{T}$ has n blue tiles and n gray tiles, this is called the *global balance condition* in [18]. By looking at the boundaries of the tiles, say $\partial\overline{T}_\alpha$, $\partial\overline{T}'_\alpha$, a tessellation $\mathcal{T}$ determines an underlying graph Γ .

Definition 2 A $\mathfrak{t}$ -graph Γ is a finite oriented connected graph embedded in $\mathcal{M}$, with vertices $V(\Gamma)$ of even valence equal or greater than 4 and edges $E(\Gamma)$, such that:

- i) $\mathcal{T}(\Gamma) \doteq \mathcal{M} \setminus \Gamma$ is a tessellation, as in Definition 1.
- ii) Each blue tile T_α is on the left side of the oriented edges of Γ .

With the above in mind, a possibly non homogeneous tessellation $\mathcal{T}$ and a $\mathfrak{t}$ -graph Γ are essentially equivalent objects, where the alternating coloring in Equation (1) corresponds to the orientation of the edges in Definition 2. In simple words, a $\mathfrak{t}$ -graph must be understood as the simplest oriented graph describing a tessellation. The tessellations arising from meromorphic functions require a more accurate notion, as follows.

Definition 3 An $\mathbf{R}$ -map $\widehat{\Gamma}$ is a finite, oriented, connected graph embedded in $\mathcal{M}$, with vertices $V(\widehat{\Gamma})$ of even valence equal or greater than 2 and edges $E(\widehat{\Gamma})$, such that:

- i) The subset of vertices of valence equal or greater than 4 is non empty.
- ii) If we forget the vertices of valence 2 of $\widehat{\Gamma}$, then we obtain a $\mathfrak{t}$ -graph Γ such that:

$$\mathcal{T}(\widehat{\Gamma}) \doteq \mathcal{T}(\Gamma) = T_1 \cup \dots \cup T_n \cup T'_1 \cup \dots \cup T'_n$$

is, set theoretically, a tessellation as in Definition 1.

- iii) The boundary $\partial\overline{T}_\alpha$ (resp. $\partial\overline{T}'_\alpha$) of each tile has $q \geq 2$ edges of $\widehat{\Gamma}$, *i.e.* the tessellation of $\widehat{\Gamma}$ is homogeneous.

The *forgetting vertices operation* in part (ii) above is as follows. We consider a vertex $z_1 = 0$ of valence 2 in $\widehat{\Gamma}$ and its two adjacent edges, thus we have $(-1, 0) \cup \{0\} \cup (0, 1)$. The operation of forgetting the vertex 0 replaces the above by a unique edge $(-1, 1)$.

After Theorem 1, the name $\mathbf{R}$ -map for $\widehat{\Gamma}$ must be understood as a coarse abbreviation of (rational) meromorphic function.

Remark 1 An $\mathbf{R}$ -map $\widehat{\Gamma}$ has two numerical attributes $n, q \geq 2$:

- n is the *degree* of $\widehat{\Gamma}$ and
- q is the *q -gonality* (the tiles of the tessellation $\mathcal{T}(\widehat{\Gamma})$ are topological q -gons).

Example 1 Figure 1.a illustrates an affine view of a $\mathfrak{t}$ -graph $\Gamma \subset \widehat{\mathbb{C}}$, whereas Figure 1.c sketches an $\mathbf{R}$ -map $\widehat{\Gamma} \subset \widehat{\mathbb{C}}$, where $\infty \in \widehat{\mathbb{C}}$ is a vertex of both graphs. In both cases, the tessellation is set theoretically the same; the difference is on the vertices of valence 2. As a consequence, the tessellation $\mathcal{T}(\Gamma)$ is non homogeneous, while $\mathcal{T}(\widehat{\Gamma})$ is homogeneous. Several tessellations $\mathcal{T}(\Gamma)$ are illustrated in [1] and [11].

A main combinatorial feature of our graphs is the following.

Definition 4 A *consistent q -labelling*

$$\mathcal{L}_c : V(\Gamma) \longrightarrow \mathbb{Z}/q, \quad q \geq 2,$$

for a $\mathfrak{t}$ -graph Γ satisfies the following conditions:

- i) For each blue tile T_α of the tessellation $\mathcal{T}(\Gamma)$, if $\{z_\iota\}$ are the vertices of its boundary $\partial \overline{T_\alpha}$, ordered with cyclic anti-clockwise sense, then the labels $\{\mathcal{L}_c(z_\iota)\} \subset \mathbb{Z}/q$ appear exactly once and with the same cyclic order as in $\mathbb{Z}/q$.
- ii) Each label $j \in \mathbb{Z}/q$ appears under $\mathcal{L}_c$ for at least one vertex $z_\iota \in V(\Gamma)$ of valence equal or greater than 4.

Remark 2 (Consistent q -labelling for $\mathbf{R}$ -maps) 1. The notion of consistent q -labelling for a $\mathfrak{t}$ -graph Γ extends to any $\mathbf{R}$ -map $\widehat{\Gamma}$ as follows: for $\widehat{\Gamma}$ all the labels of $\mathbb{Z}/q$ appear on the vertices of each blue tile T_α , since all the tiles of $\mathcal{T}(\widehat{\Gamma})$ are q -gons.

2. On the other hand, for Γ usually some labels of $\mathbb{Z}/q$ are hidden in the boundary of each blue tile, since the tiles of the tessellation $\mathcal{T}(\Gamma)$ can be ρ -polygons, for $2 \leq \rho \leq q$, as we will show in Example 2 and Figure 1. By abuse of notation, we use the notion of consistent q -labelling for $\mathfrak{t}$ -graphs and $\mathbf{R}$ -maps.

3. Note that not all $\mathbf{R}$ -maps have consistent q -labellings. See for instance figure 10 of [18]. The figure depicts an $\mathbf{R}$ -map with degree 4 (number of tiles of each color), 6-gonality (each tile is a 6-gon), and a labelling in $\mathbb{Z}/6$. However, all the vertices labelled 5 have valence 2, thus,

- the labelling is not a consistent 6-labelling,
- the vertices labelled 5 are regular points and 5 is a fake critical value.

By forgetting the vertices labelled 5, we obtain another $\mathbf{R}$ -map with degree 4 and 5-gonality, renaming label 6 to label 5 provides a consistent 5-labelling.

2.1 Schwarz–Klein’s algorithm for the construction of tessellations

We follow the classical works of H. A. Schwarz [29], F. Klein [16], [7], R. Nevanlinna [27] ch. XI §2.

Schwarz–Klein’s algorithm.

Let $R : M \longrightarrow \widehat{\mathbb{C}}$ be a meromorphic function of degree $n \geq 2$. Recall the Riemann–Hurwitz formula

$$2g - 2 = -2n + \sum_{z_\iota} (\mu_\iota - 1),$$

where $\{z_\iota\}$ are the singular points of R with ramification orders $\mu_\iota \geq 2$. In particular, if m is the number of critical points of R , then $2 + 2g \leq m \leq 2n + 2g - 2$.

Step 1. Compute the critical points of R ,

$$\mathcal{CP}_R = \{z_1, \dots, z_m\} \subset M$$

and the critical values of R ,

$$\mathcal{CV}_R = \{w_1, \dots, w_j, \dots, w_q\} \subset \widehat{\mathbb{C}}, \quad j \in 1, \dots, q, \quad 2 + 2g \leq q \leq m.$$

Step 2. Construct an oriented Jordan path

$$\gamma \subset \widehat{\mathbb{C}} \text{ running through } \{w_1, \dots, w_q\},$$

here the cyclic subindices $j \in \mathbb{Z}/q$ of the critical values are provided by γ . We have a trivial tessellation

$$\mathcal{T}(\gamma) = \widehat{\mathbb{C}} \setminus \gamma = T \cup T' \tag{2}$$

with topological q -gons as tiles, T is blue (in the left side of γ) and T' is gray. Moreover, γ is a cyclic graph with q vertices $\{w_j\} = \mathcal{CV}_R$ and q edges (the respective segments $\overline{w_l w_{l+1}}$, $\overline{w_q w_1}$ of γ).

Step 3. Compute the inverse image of γ ,

$$\Gamma = R^{-1}(\gamma) \subset M. \quad (3)$$

More accurately, the pullback graph

$$\widehat{\Gamma} = R^* \gamma \quad (4)$$

is well defined. By definition, a *cocritical point* $\zeta_\kappa \in \mathcal{Cc}_R$ of R satisfies that $R(\zeta_\kappa)$ is a critical value but ζ_κ is a regular point of R . The vertices of $\widehat{\Gamma}$ are

$$V(\widehat{\Gamma}) = \underbrace{\{\text{critical points } \mathcal{CP}_R\}}_{\text{even valence } \geq 4} \cup \underbrace{\{\text{cocritical points } \mathcal{Cc}_R\}}_{\text{valence } = 2}. \quad (5)$$

The edges $E(\widehat{\Gamma})$ are the respective segments in the R-map $\widehat{\Gamma} = R^* \gamma$. Set theoretically $\Gamma = \widehat{\Gamma}$, however they are isomorphic graphs if and only the cocritical point set $\mathcal{Cc}_R$ is empty.

Step 4. The tessellation determined by R and γ is

$$\mathcal{T}_\gamma(R) = M \setminus \widehat{\Gamma} = \underbrace{T_1 \cup \dots \cup T_n}_{\text{blue tiles}} \cup \underbrace{T'_1 \cup \dots \cup T'_n}_{\text{gray tiles}}, \quad n \geq 2.$$

Step 5. The cocritical points $\mathcal{Cc}_R$ play a crucial role, hence the tiles of $\mathcal{T}(\widehat{\Gamma})$ are topological q -gons. Finally, we add labels to the vertices of γ with the labelling map

$$\mathcal{L}_\gamma : V(\gamma) \longrightarrow \mathbb{Z}/q, \quad w_j \longmapsto j. \quad (6)$$

The pullback of the labels determines a *consistent q -labelling* for $\widehat{\Gamma}$

$$\begin{aligned} R^* \mathcal{L}_\gamma : \mathcal{CP}_R \cup \mathcal{Cc}_R &\longrightarrow \mathbb{Z}/q \\ z_l &\longmapsto \mathcal{L}_\gamma(R(z_l)). \end{aligned} \quad (7)$$

Summing up, the input of the Schwarz–Klein’s algorithm is the pair $(M, R, \mathcal{L}_\gamma)$. The (equivalent) outputs are:

- The homogeneous tessellation $\mathcal{T}_\gamma(R)$ of M with $2n$ tiles, which are topological q -gons having alternating colors.
- The R-map $\widehat{\Gamma} = R^* \gamma \subset M$ with a consistent q -labelling $R^* \mathcal{L}_\gamma$.

Example 2 (Tessellation of a non generic meromorphic function) Let

$$R(z) : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}, \quad z \longmapsto z(z^2 - 1)(z^2 - 4)/(z - 3)$$

be a non generic³ meromorphic function of degree 5, as a straightforward computation shows, there are six real critical points $\{z_1, \dots, z_5, \infty\}$, with five real critical values and infinity. Hence

³A meromorphic function $R(z)$ in $\widehat{\mathbb{C}}$ of degree n is *generic* when it has $2n - 2$ distinct critical points.

the choice of $\gamma = \mathbb{R} \cup \{\infty\}$ is suitable. Figure 1.a sketches the affine view of Γ , the finite critical points and its tessellation $\mathcal{T}(\Gamma)$. The path γ induces the cyclic order $w_1 < w_2 < w_3 < w_4 < w_5 < w_6 = \infty$. By Equation (6), there is a labelling for the critical values of $R(z)$,

$$\mathcal{L}_\gamma : \mathcal{CV}_R \longrightarrow \mathbb{Z}/6, \quad w_j \longmapsto j,$$

where by simplicity the subindex j coincides with a cyclic order in $\mathcal{CV}_R$. Without loss of generality, we define that $\mathcal{L}_\gamma(\infty) = 6$.

The pullback of $\mathcal{L}_\gamma$ under R determines a consistent 6-labelling of the critical and cocritical points of R , that is

$$\mathcal{L}_c \doteq R^* \mathcal{L}_\gamma : \mathcal{CP}_R \cup \mathcal{CC}_R \longrightarrow \mathbb{Z}/6, \quad z_l \longmapsto \mathcal{L}_\gamma(R(z_l)).$$

Figure 1.c sketches the affine view of $\hat{\Gamma} = R^* \gamma$, its homogeneous tessellation $\mathcal{T}(\hat{\Gamma})$ and the consistent 6-labelling $\mathcal{L}_c : V(\hat{\Gamma}) \longrightarrow \mathbb{Z}/6$, with $\mathcal{L}_c(\infty) = 6$. In particular, the labelling $\mathcal{L}_c$ of Γ is computed as

$$\begin{aligned} z_1 \mapsto R(z_1) = w_5 \mapsto \mathcal{L}_\gamma(w_5) = 5, & \quad z_4 \mapsto R(z_4) = w_2 \mapsto \mathcal{L}_\gamma(w_2) = 2, \\ z_2 \mapsto R(z_2) = w_3 \mapsto \mathcal{L}_\gamma(w_3) = 3, & \quad z_5 \mapsto R(z_5) = w_1 \mapsto \mathcal{L}_\gamma(w_1) = 1, \\ z_3 \mapsto R(z_3) = w_4 \mapsto \mathcal{L}_\gamma(w_4) = 4, & \quad z_6 = \infty \mapsto R(z_6) = \infty \mapsto \mathcal{L}_\gamma(w_6) = 6. \end{aligned}$$

The q -gonality of $\hat{\Gamma} = R^* \gamma$ is 6, *i.e.* the tiles are topological 6-gons.

The output of the Schwarz–Klein’s algorithm applied to R is $\mathcal{T}_\gamma(R)$, or $\hat{\Gamma}$ with a labelling $\mathcal{L}_c$ with values in $\mathbb{Z}/6$.

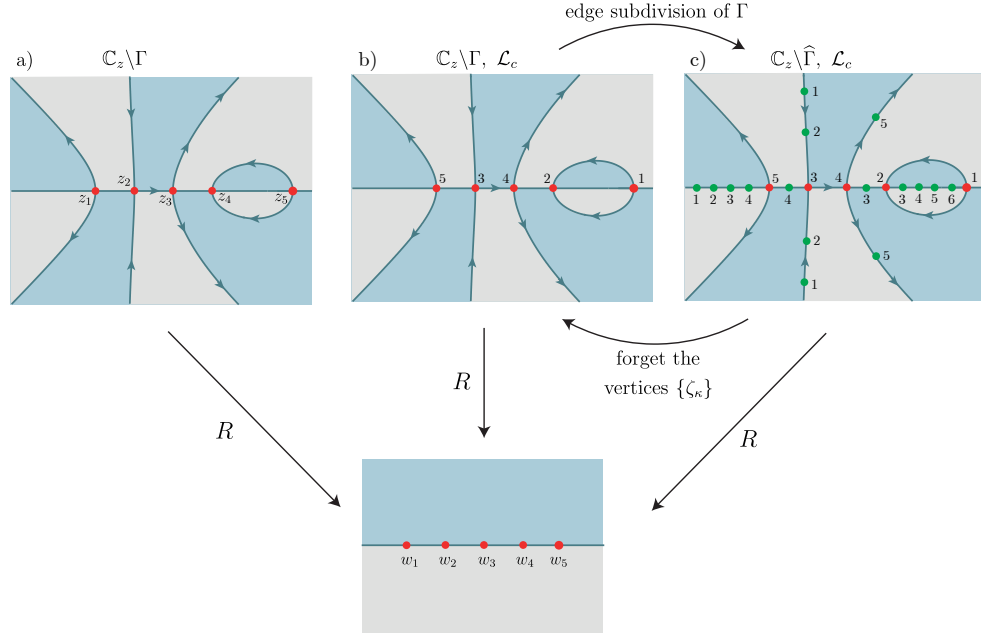


Fig. 1 Affine view of the tessellation of a non generic meromorphic function $R(z)$ of degree 5. a) The $\mathbf{t}$ -graph $\Gamma = R^{-1}(\mathbb{R} \cup \{\infty\})$ and its non homogeneous tessellation $\mathcal{T}(\Gamma)$. b) The consistent 6-labelling $\mathcal{L}_c : V(\Gamma) \longrightarrow \mathbb{Z}/6$, where $\mathcal{L}(\infty) = 6$. c) The $\mathbf{R}$ -map $\hat{\Gamma} = R^* \gamma$ and its homogeneous tessellation $\mathcal{T}(\hat{\Gamma})$: it has a consistent 6-labelling $R^* \mathcal{L}_c$, each tile is a 6-gon, with vertices at the (red) critical points $\mathcal{CP}_R$, the point $w_6 = \infty$ (which has label 6), and the (green) cocritical points $\mathcal{CC}_R$.

As a second part of the example, assume that our departure point is the $\mathfrak{t}$ -graph Γ describing the non homogeneous tessellation $\mathcal{T}(\Gamma)$ in Figure 1.a, with $10 = 2n$ tiles. We want to ask about the existence of a meromorphic function R of degree $n = 5$, which topologically realizes the tessellation of Γ .

As a main difficulty, note that the tiles of $\mathcal{T}(\Gamma)$ are ρ -gons, for $\rho = 2, 3, 4$, providing a lower bound for q , on the other hand, the Riemann–Hurwitz formula provides an upper bound; we must therefore search for a consistent q -labelling with

$$\max\{\rho\} = 4 \leq q \leq 8 = 2n + 2g - 2.$$

Further, since $g = 0$ and noting that condition (ii) of Definition 4 implies $q \leq \#\{\text{vertices of valence } \geq 4\} = 6$, or search reduces to $4 \leq q \leq 6$.

The key assumption is whether we know of consistent q -labellings for the vertices of Γ . In Figure 1.b, a consistent 6-labelling is shown for the $\mathfrak{t}$ -graph. In order to construct an R -map that encodes a topological branched covering $\mathcal{R}$ over $\hat{\mathbb{C}}$, we proceed by an edge subdivision operation in Γ , as in Figure 1.c. Finally, the existence of a complex structure, making it a meromorphic function is assertion (2) of Theorem 1.

As it turns out, for this particular non homogeneous tessellation $\mathcal{T}(\Gamma)$ there are also consistent q -labellings for $q = 4$, and $q = 5$: Figure 2 shows one for $q = 4$ and two for $q = 5$; each gives rise to a different meromorphic function R .

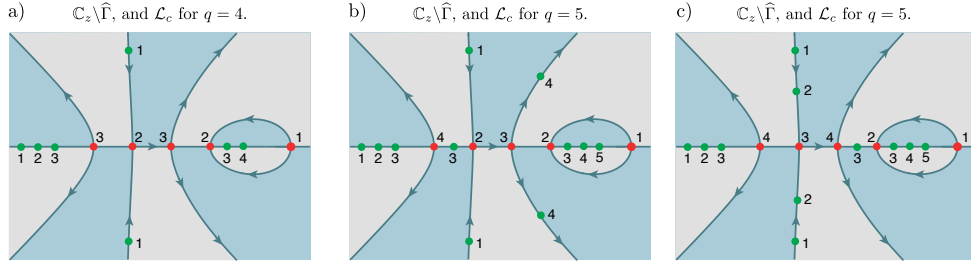


Fig. 2 Consistent q -labellings for $q = 4$, and 5 of the same $\mathfrak{t}$ -graph of Figure 1.a which originates from $R(z)$ in Example 2. a) A consistent 4-labelling $\mathcal{L}_c : V(\Gamma) \rightarrow \mathbb{Z}/4$, where $\mathcal{L}(\infty) = 4$. b) A consistent 5-labelling $\mathcal{L}_c : V(\Gamma) \rightarrow \mathbb{Z}/5$, where $\mathcal{L}(\infty) = 5$. c) A topologically different consistent 5-labelling $\mathcal{L}_c : V(\Gamma) \rightarrow \mathbb{Z}/5$, where $\mathcal{L}(\infty) = 5$. Each case produces a different R -map $\hat{\Gamma} = R^*\gamma$ with a homogeneous tessellation $\mathcal{T}(\hat{\Gamma})$. Each tile is a q -gon, with vertices at the (red) critical points $\mathcal{CP}_R$, the point ∞ with label q , and the (green) cocritical points $\mathcal{Cc}_R$.

Figure 3.a–b in [11] illustrates the tessellation $\mathcal{T}(\Gamma) \subset \mathbb{S}^2$ of a $\mathfrak{t}$ -graph that admits consistent q -labellings for $q = 3$ and 4 .

3 A converse for the Schwarz–Klein’s algorithm

Let $\Gamma \subset \mathcal{M}$ be a $\mathfrak{t}$ -graph, where $\mathcal{M}$ is a compact oriented C^1 surface and such that $\mathcal{T}(\Gamma) = \mathcal{M} \setminus \Gamma$ is a tessellation, as in Definition 1.

Proof of Theorem 1 assertion (1). Follows directly from the Schwarz–Klein’s algorithm. Let us comment that this assertion assumes that M is a Riemann surface, whereas assertion (2) will work on $\mathcal{M}$:

Remark 3 For $q = 2$ critical values, the only functions that appear are meromorphic functions of degree $n = 2$ on $\widehat{\mathbb{C}}$ with two critical points.

For $q = 3$ critical values the pair $(\Gamma, \mathcal{L}_c)$ determines a unique complex structure on $\mathcal{M}$, see [15, p. 4] for a historical comment. This fortunate property is key for the theory of Belyi’s functions and dessins d’enfants.

For $q \geq 4$, clearly, a pair $(\Gamma, \mathcal{L}_c)$ determines families of non biholomorphic complex structures (M, J) and meromorphic functions R , see Example 5.1.

Proof of Theorem 1 assertion (2). We perform several steps.

Step 1. By using the consistent q -labelling $\mathcal{L}_c$ and edge subdivision operation for Γ , we get an associated R-map $\widehat{\Gamma}$ in $\mathcal{M}$ with a consistent q -labelling $\mathcal{L}_c : V(\widehat{\Gamma}) \rightarrow \mathbb{Z}/q$, as follows.

Edge subdivision operation. Let $\overline{z_\iota z_\sigma}$ an edge of Γ with labels $\mathcal{L}_c(z_\iota)$ and $\mathcal{L}_c(z_\sigma)$.

If $\mathcal{L}_c(z_\sigma) - \mathcal{L}_c(z_\iota) = 1 \pmod{q}$, then $\overline{z_\iota z_\sigma}$ is an edge of $\widehat{\Gamma}$.

If $\mathcal{L}_c(z_\sigma) - \mathcal{L}_c(z_\iota) = \nu + 1 \geq 2 \pmod{q}$, then we consider ν new vertices in the original edge $\overline{z_\iota z_\sigma}$, obtaining $\nu + 1$ edges

$$\overline{z_\iota \zeta_1}, \overline{\zeta_1 \zeta_2}, \dots, \overline{\zeta_\nu z_\sigma}$$

of $\widehat{\Gamma}$. Moreover, the labels of these new vertices of valence 2 of $\widehat{\Gamma}$ are

$$\mathcal{L}_c(z_\iota) = h, \mathcal{L}_c(\zeta_1) = h + 1, \dots, \mathcal{L}_c(\zeta_\nu) = h + \nu, \mathcal{L}_c(z_\sigma) = h + \nu + 1; \pmod{q}.$$

Figure 1.b–c illustrates the edge subdivision operation.

Step 2. We consider the circle $\gamma = \mathbb{R} \cup \{\infty\} \subset \widehat{\mathbb{C}}$ furnished with vertices

$$V(\gamma) = \{1, 2, \dots, k-1, \infty\}$$

and the respective q segments as edges $E(\gamma)$. Thus γ determines a trivial tessellation (2), say

$$\widehat{\mathbb{C}} \setminus \gamma = \mathbb{H}_+^2 \cup \mathbb{H}_-^2,$$

where as usual $\mathbb{H}_+^2$ denotes the open upper half plane. The labelling of the vertices of γ is

$$\mathcal{L}_\gamma : \underbrace{\{1, 2, \dots, q-1, \infty\}}_\varsigma \subset \widehat{\mathbb{C}} \longrightarrow \mathbb{Z}/q, \quad \begin{cases} \varsigma \longmapsto \varsigma \\ \infty \longmapsto q. \end{cases}$$

Remark 4 Throughout the work $(\widehat{\mathbb{C}}, \partial/\partial w)$ denotes the Riemann sphere furnished with the holomorphic vector field $\partial/\partial w$, see Figure 3. By abuse of notation, this pair must be understood as the euclidean or flat Riemannian metric, with a singularity at $\infty \in \widehat{\mathbb{C}}$. The concepts of euclidean segments and trajectories of the real vector field

$$\Re \left(e^{i\theta} \frac{\partial}{\partial w} \right) \doteq \cos(\theta) \frac{\partial}{\partial x} + \sin(\theta) \frac{\partial}{\partial y}, \quad \theta \in [0, 2\pi],$$

in $\widehat{\mathbb{C}}$ are used in an equivalent way. On Riemann surfaces, there exists a fruitful relationship between complex analytic functions and vector fields, it is described as the “Dictionary” [2, prop. 2.5].

Since each tile T_α or T'_α of $\mathcal{T}(\widehat{\Gamma})$ is an open Jordan domain, there exist C^1 diffeomorphisms

$$\Psi_{\alpha+} : T_\alpha \subset \mathcal{M} \longrightarrow \mathbb{H}_+^2 \subset \left(\widehat{\mathbb{C}}, \frac{\partial}{\partial w} \right), \quad \Psi_{\alpha-} : T'_\alpha \subset \mathcal{M} \longrightarrow \mathbb{H}_-^2 \subset \left(\widehat{\mathbb{C}}, \frac{\partial}{\partial w} \right), \quad (8)$$

which extend continuously to the boundary $\partial \overline{T_\alpha}, \partial \overline{T'_\alpha}$. That is, $\Psi_{\alpha+} : \partial \overline{T_\alpha} \longrightarrow \gamma$ is a homeomorphism and the same property holds for $\Psi_{\alpha-}$.

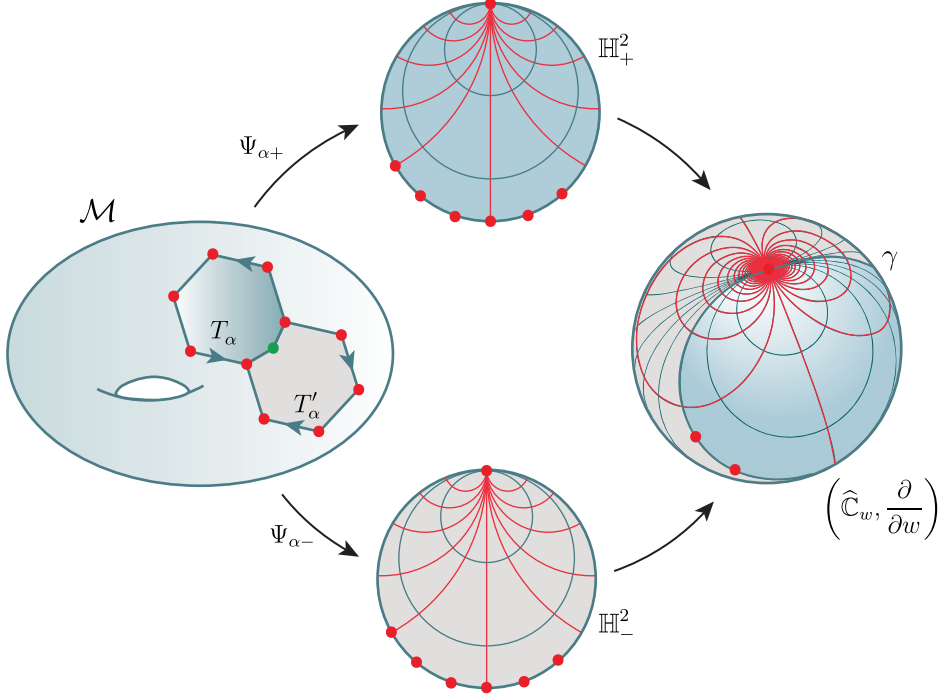


Fig. 3 Two topological q -gons T_1, T'_1 are mapped to the half planes $\mathbb{H}_+^2, \mathbb{H}_-^2$. We sketch the case $q = 7$, the critical points are red points in $\mathcal{M}$, a cocritical point is colored green.

The geometrical structure provided by the holomorphic vector field $\frac{\partial}{\partial w}$ on $\widehat{\mathbb{C}}$ is as follows. Recall that the real vector fields

$$\Re \left(\frac{\partial}{\partial w} \right) \doteq \frac{\partial}{\partial x}, \quad \Im \left(\frac{\partial}{\partial w} \right) \doteq \frac{\partial}{\partial y}$$

are real analytic on $\widehat{\mathbb{C}}$ with a “dipole” at $w = \infty$ as the unique singularity. The usual almost complex structure J of $\widehat{\mathbb{C}}$ is provided by

$$J \left(\frac{\partial}{\partial x} \right) = \frac{\partial}{\partial y}, \quad J \left(\frac{\partial}{\partial y} \right) = -\frac{\partial}{\partial x}.$$

Recall that, on a smooth surface $\mathcal{M}$ every almost complex structure is integrable, *i.e.* determines a complex structure, see [17] ch. XI §2 for details.

The pullback complex structure on T_α is

$$J_{\alpha+}((\Psi_{\alpha+})^* \frac{\partial}{\partial x}) = (\Psi_{\alpha+})^* \frac{\partial}{\partial y}, \quad J_{\alpha+}((\Psi_{\alpha+})^* \frac{\partial}{\partial y}) = -(\Psi_{\alpha+})^* \frac{\partial}{\partial x}.$$

Summing up, the pair $(T_\alpha, J_{\alpha+})$ is an open Riemann surface with holomorphic chart $\Psi_{\alpha+}$ as in (8). Analogously, each $(T'_\alpha, J_{\alpha-})$ is an open Riemann surface.

Step 3. By using the labelled graphs $(\widehat{\Gamma}, \mathcal{L}_c)$ and $(\gamma, \mathcal{L}_\gamma)$, we construct a local homeomorphism

$$\mathcal{R} : \mathcal{M} \setminus V(\widehat{\Gamma}) \longrightarrow \widehat{\mathbb{C}} \setminus \{1, \dots, q-1, \infty\},$$

defined in the interior of the tiles as $\Psi_{\alpha+}$ or $\Psi_{\alpha-}$. Hence $\mathcal{R}$ satisfies that $\mathcal{R}(T_\alpha) = \mathbb{H}_+^2$ and $\mathcal{R}(T'_\alpha) = \mathbb{H}_-^2$.

Let $\zeta_0 \in E(\widehat{\Gamma}) \subset \mathcal{M}$ be a point in an edge of $\widehat{\Gamma}$ but not a vertex of it, we want to construct a holomorphic chart for ζ_0 . There exist a unique trajectory of the C^1 vector field $(\Psi_{\alpha+})^* \frac{\partial}{\partial y}$ in T_α , say the vertical line $\{x = x_0\} \subset \mathbb{H}_+^2$, such that ζ_0 is its unique extreme point. Figure 3 illustrates this behavior. We define

$$\mathcal{R}(\zeta_0) = x_0 + 0i \in \mathbb{R} \setminus \{1, \dots, q-1, \infty\} \subset \widehat{\mathbb{C}}.$$

Moreover, since ζ_0 is in the boundary of a gray tile, say T'_α , then there exists a unique trajectory of the C^1 vector field $(\Psi_{\alpha-})^* \frac{\partial}{\partial y}$ in T'_α , given by the vertical line $\{x = x_0\} \subset \mathbb{H}_+^2$, with ζ_0 is its unique extreme point. Hence, $\mathcal{R}$ is well defined and continuous at ζ_0 .

Step 4. We construct a holomorphic chart that covers ζ_0 . Let ϵ the minimal euclidean distance of $\mathcal{R}(\zeta_0) = x_0$ to the vertices of γ , thus $\epsilon = \min\{|x_0 - m| \mid m \in 1, \dots, q-1\} > 0$. We denote by $D(x_0, \epsilon) \subset \mathbb{C}$ the complex disk, with center $x_0 \in \mathbb{C}$ and radius ϵ . Define a continuous chart

$$\Psi_{\zeta_0}^{-1} : D(x_0, \epsilon) \subset \mathbb{C} \longrightarrow \mathcal{V}(\zeta_0) \subset \mathcal{M} \quad (9)$$

as follows:

- i) If $x + i0 \in D(x_0, \epsilon) \subset \mathbb{C} \cap \mathbb{R}$, then $\Psi_{\zeta_0}^{-1}(x)$ is a orientation preserving isometry map from the euclidean segment $(x_0 - \epsilon, x_0 + \epsilon)$ to the edge of $\widehat{\Gamma}$ that contains ζ_0 .
- ii) If $x + iy \in D(x_0, \epsilon) \subset \mathbb{C} \cap \mathbb{H}_+^2$, then $\Psi_{\zeta_0}^{-1}(x + iy) \in T_\alpha$ is the unique point such that under the real flow of the real vector field $(\Psi_{\alpha+})^* \frac{\partial}{\partial y}$ by time $-y$ arrives to $\Psi_{\zeta_0}^{-1}(x)$.
- iii) If $x + iy \in D(x_0, \epsilon) \subset \mathbb{C} \cap \mathbb{H}_-^2$, then $\Psi_{\zeta_0}^{-1}(x + iy) \in T'_\alpha$ is the unique point such that under the real flow of the real vector field $(\Psi_{\alpha-})^* \frac{\partial}{\partial y}$ by time y arrives to $\Psi_{\zeta_0}^{-1}(x)$.

A priori, the map $\Psi_{\zeta_0}^{-1}$ is not C^1 at the edges of $\widehat{\Gamma}$, when you consider the original C^1 structure on $\mathcal{M}$.

According to the Equations (8) and (9), when the compositions $\Psi_{\alpha+} \circ \Psi_{\zeta_0}^{-1}$, $\Psi_{\alpha-} \circ \Psi_{\zeta_0}^{-1}$ are well defined, they are the restrictions of translations in $\mathbb{C}$. Hence, the collection

$$\{\Psi_{\alpha+}, \Psi_{\alpha-}, \Psi_{\zeta_0} \mid \alpha \in 1, \dots, n, \zeta_0 \in E(\widehat{\Gamma}) \setminus V(\widehat{\Gamma})\},$$

is a holomorphic atlas for $\mathcal{M} \setminus V(\widehat{\Gamma})$. We denote the resulting open Riemann surface as M° .

Step 5. The construction of a holomorphic map from

$$R^\circ : M^\circ \longrightarrow \widehat{\mathbb{C}}$$

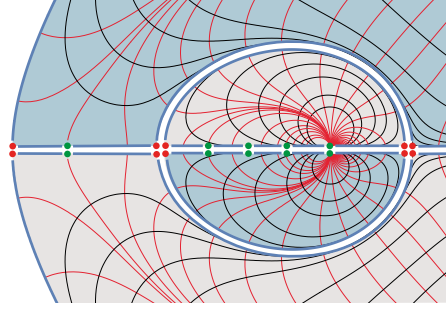


Fig. 4 Zoom of Figure 1 illustrates four tiles before the gluing process. The tiles are provided with the real trajectories of the vector fields $\Psi_{\alpha\pm}^* \frac{\partial}{\partial x}$ in black and $\Psi_{\alpha\pm}^* \frac{\partial}{\partial y}$ in red. Moreover, the critical points are red, the cocritical points are green. A simple pole of $R(z)$ appears as cocritical point.

is immediate. Consider the blue tile T_1 and the point $z_0 \in T_1$ such that $\Psi_{1+}(z_0) = i$. We define $R^\circ = \Psi_{1+}$ in T_1 . If T'_β is any gray tile adjacent to T_1 , then we extend our function Ψ_{1+} by analytic continuation, that is

$$R^\circ : T_1 \cup T'_\beta \cup \overline{z_j z_l} \longrightarrow \widehat{\mathbb{C}}, \quad \mathfrak{z} \longmapsto \begin{cases} \Psi_{1+}(\mathfrak{z}) & \mathfrak{z} \in T_1 \\ \Psi_{\beta-}(\mathfrak{z}) & \mathfrak{z} \in T'_\beta, \end{cases}$$

where $\cup \overline{z_j z_l}$ denotes the interior of the edges of $\widehat{\Gamma}$ in $\partial T_1 \cap T'_\beta$. In fact, with the above definition Ψ_{1+} extends holomorphically to M° , it produces a holomorphic function R° , as we asserted.

Step 6. Using the Riemann's extension theorem, we have that the vertices $V(\widehat{\Gamma})$ are conformal punctures of M° , since the above atlas can be extended to $V(\widehat{\Gamma})$. Thus M is a compact Riemann surface. Let

$$R : M \longrightarrow \widehat{\mathbb{C}}$$

be the analytic extension of R° to the vertices $V(\widehat{\Gamma})$. We describe the local behavior of R at each vertex. Note that, R° near z_l is topologically equivalent to

$$\{z \longmapsto z^{\nu/2}\}, \quad \text{where } \nu = 2 \text{ or } \nu \geq 4 \text{ and even,}$$

means the valence of the vertex⁴ z_l , as a consequence:

If $z_l \in V(\widehat{\Gamma})$ has label $j \in \{1, \dots, q-1\}$ then z_l is

$$\begin{cases} \text{a critical point of } R, \text{ with finite critical value, and ramification index } \nu/2 \text{ when } \nu \geq 4, \\ \text{a cocritical point of } R \text{ when } \nu = 2. \end{cases}$$

If $z_l \in V(\widehat{\Gamma})$ has label q then z_l is

$$\begin{cases} \text{a pole of order } \nu/2 \text{ of } R \text{ when } \nu \geq 4, \\ \text{a simple pole of } R \text{ when } \nu = 2. \end{cases}$$

⁴Recall that the vertices of $\widehat{\Gamma}$ split into two families: those of valence 2 and those of valence ≥ 4 , denoted as $\{\zeta_\kappa\}$ and $\{z_\iota\}$ respectively (see Step 1 and Step 3 of Schwarz–Klein's algorithm in Section 2.1). In what follows we will use the notation $\{z_\iota\}$ for both families.

The construction of the meromorphic function R and assertion (2) of Theorem 1 are done.

Corollary 1 1) A $\mathfrak{t}$ -graph Γ in $\mathcal{M}$, provided with a consistent labelling $\mathcal{L}_c$, determines a ramified cover $\mathcal{R} : \mathcal{M} \rightarrow \widehat{\mathbb{C}}$.

2) Conversely, a C^1 ramified cover $\mathcal{R}$ determines a (non unique) meromorphic function $R : M \rightarrow \mathbb{C}_w$.

3) Let $\mathcal{T}_\gamma(R)$ be a tessellation of M , from a meromorphic function R and a Jordan path γ . If the tiles T_α, T'_α are adjacent along an edge $\overline{z_j z_i}$ of $\widehat{\Gamma}$, then there is a branch of the inverse function

$$R^{-1} : (\widehat{\mathbb{C}} \setminus \gamma) \cup R(\overline{z_j z_i}) \rightarrow T_\alpha \cup \overline{z_j z_i} \cup T'_\alpha \subset M,$$

where $(\widehat{\mathbb{C}} \setminus \gamma) \cup R(\overline{z_j z_i})$ is a maximal domain of univalence. $\square$

Some comments are in order.

The function R in Theorem 1 assertion (2) is not unique, due to several choices:

- i) The complex structure on M depends on the choice of the critical values $\{1, \dots, q-1, \infty\} \subset \widehat{\mathbb{C}}$
- ii) For any q distinct critical values $\{w_1, \dots, w_q\} \subset \mathbb{C}_w$, the construction with the same technique remains simple when γ is a polygonal path in $\widehat{\mathbb{C}}$, as we will show in Corollary 2.
- iii) For fixed $\{w_1, \dots, w_q\}$ the change of γ was studied by Habsch [14], see also [19, § 5.4]; both in the case $M = \widehat{\mathbb{C}}$. However, note that if γ is allowed to vary by an isotopy relative to the q distinct critical values, then the function R is unchanged.

3.1 Relationship between analytical and combinatorial structures

As a matter of record, let us consider the following.

- A homogeneous tessellation $\mathcal{T}_\gamma(R)$, or equivalently an analytical triple (M, R, γ) , denotes a meromorphic function $R : M \rightarrow \widehat{\mathbb{C}}$, on a compact Riemann surface, and a cyclic order $\mathcal{L}_\gamma$ of its q critical values $\mathcal{CV}_R$ of R .
- A pair $(\widehat{\Gamma}, \mathcal{L}_c)$, where $\widehat{\Gamma} \subset \mathcal{M}$ is an $\mathfrak{R}$ -map (on a C^1 compact oriented surface $\mathcal{M}$), with vertices of even valence ≥ 2 , and a consistent q -labelling $\mathcal{L}_c : V(\widehat{\Gamma}) \rightarrow \mathbb{Z}/q$.
- A pair $(\Gamma, \mathcal{L}_c)$, where $\Gamma \subset \mathcal{M}$ is a $\mathfrak{t}$ -graph, with vertices of even valence ≥ 4 , and a consistent q -labelling $\mathcal{L}_c : V(\Gamma) \rightarrow \mathbb{Z}/q$.
- $\mathcal{T} = \mathcal{T}(\Gamma) = \mathcal{M} \setminus \Gamma$ is a tessellation.

Note that only $\mathcal{T}_\gamma(R)$ belongs to the complex analytic category. The q -labellings of $\widehat{\Gamma}$ and Γ coincide on the vertices of Γ . We have the following correspondences:

$$\begin{array}{ccccc}
& & & \textcircled{5} & \\
& & & \longleftarrow & \mathcal{T}_\gamma(R). \\
& & (\widehat{\Gamma}, \mathcal{L}_c) & & \\
& \textcircled{3} \updownarrow \textcircled{4} & & & \\
& (\Gamma, \mathcal{L}_c) & \xrightarrow{\textcircled{6}} & & \\
\mathcal{T}(\Gamma) & \xleftarrow{\textcircled{1}} \Gamma & \xleftarrow{\textcircled{2}} & &
\end{array} \tag{10}$$

Map $\textcircled{1}$ is an equivalence, by definition.

Map $\textcircled{2}$ forgets the labels $\{\mathcal{L}_c(z_i)\}$ of the vertices of Γ .

Map $\textcircled{3}$ forgets the vertices $\{\zeta_\kappa\} \subset \widehat{\Gamma}$ of valence 2 and their labels $\{\mathcal{L}_c(\zeta_\kappa)\}$.

Map $\textcircled{4}$ is the edge subdivision of Γ , that introduces vertices of valence 2, according to the consistent q -labelling $\mathcal{L}_c$ of $V(\Gamma)$.

Map $\textcircled{5}$ is assertion (1) in Theorem 1, where $\widehat{\Gamma} = R^*\gamma$ and the consistent q -labelling $\mathcal{L}_c : V(\widehat{\Gamma}) \rightarrow \mathbb{Z}/q$ is the pullback $R^*\mathcal{L}_\gamma$ of the cyclic order $\mathcal{L}_\gamma$ of the critical values of R .

Map $\textcircled{6}$ is assertion (2) in Theorem 1.

The tessellations $\mathcal{T}(\widehat{\Gamma})$ and $\mathcal{T}(\Gamma)$ coincide, set theoretically.

The construction of R in the proof of Theorem 1 produces critical values in $\mathbb{R} \cup \{\infty\}$. We introduce now a more flexible construction.

Corollary 2 (Construction of meromorphic functions by surgery) *Let $\overline{\mathcal{P}} \subset (\widehat{\mathbb{C}}, \partial/\partial w)$ be a closed polygon with vertices $\{w_1, \dots, w_q\} \subset \widehat{\mathbb{C}}$ and oriented euclidean segments as edges, where $\mathcal{P}$ is on the left side of the edges. Assume that $\mathcal{P}$ and its complement $\overline{\mathcal{P}}^c$ are open Jordan domains. Consider a collection of $2n$ tiles*

$$T_\alpha = \left(\mathcal{P}, \frac{\partial}{\partial w}\right), \quad T'_\alpha = \left(\overline{\mathcal{P}}^c, \frac{\partial}{\partial w}\right), \quad \alpha \in \{1, \dots, n \geq 2\}.$$

Assume that M is a compact Riemann surface obtained by gluing together closed tiles $\overline{T}_\alpha$ and T'_β , along an edge by euclidean isometries. Each vertex $z_i \in M$ (arising from the gluing of vertices in the closed tiles) has cone angle $2\pi\nu_i$, $\nu_i \in \mathbb{N}$. Then M determines

- i) a meromorphic function $R : M \rightarrow \widehat{\mathbb{C}}$ of degree n with tessellation $\mathcal{T}_\gamma(R)$, where $\gamma = \partial\overline{\mathcal{P}}$, and*
- ii) a pullback meromorphic vector field $X = R^* \frac{\partial}{\partial w}$ on M .*

The case $n = \infty$ gives origin to complex analytic functions (not necessarily meromorphic) on non compact Riemann surfaces, see [3].

Proof The boundaries of the tiles determine an R -map on M . Note that, the alternating color condition for the tiles of $\mathcal{T}_\gamma(R)$ is implicit in the hypothesis. The set of points $\{w_1, \dots, w_q\}$ contains the critical value set $\mathcal{CV}_R \neq \emptyset$. If no minimality condition is assumed on the edges of $\overline{\mathcal{P}}$, then additional subdivision of an edge produces regular points and fake critical values. Furthermore, a straight line segment through ∞ is allowed as an edge in $\partial\overline{\mathcal{P}}$. The pullback of $\partial/\partial w$ under $R(z)$ is $(1/R'(z)) \frac{\partial}{\partial z}$, this analytic expression makes sense on any Riemann surface. $\square$

Corollary 3 *Let $R : M \longrightarrow \widehat{\mathbb{C}}$ be a meromorphic function. Topologically the tessellation $\mathcal{T}_\gamma(R)$ only depends on the cyclic order $\mathcal{L}_\gamma$, given to the critical values $\{w_1, \dots, w_q\}$ of R .* $\square$

In other words, the tessellation $\mathcal{T}_\gamma(R)$ is topologically independent on the particular shape of γ (as long as γ is in the same isotopy class relative to the q distinct critical points). There are $q!$ cyclic orders $\mathcal{L}_\gamma$ for the critical values $\{w_1, \dots, w_q\}$. Note that for $q \geq 4$, up to Möbius transformation, the usual choice for the critical values is $\{0, 1, \infty, w_4, \dots, w_q\}$. A complete discussion is to appear in [3].

4 Examples

Given integers $\mathbf{p}, \mathbf{q}, \mathbf{r} \geq 2$, let $G_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ be the group with generators and relations

$$\{g_1, g_2, g_3 \mid g_1^2 = g_2^2 = g_3^2 = 1, (g_1 g_2)^{\mathbf{p}} = (g_2 g_3)^{\mathbf{q}} = (g_3 g_1)^{\mathbf{r}} = 1\}.$$

Then $G_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ can be represented as a group of isometries of one of the 2-dimensional geometries:

$$\frac{1}{\mathbf{p}} + \frac{1}{\mathbf{q}} + \frac{1}{\mathbf{r}} \begin{cases} > 1 & \text{spherical } \mathbb{S}^2, \\ = 1 & \text{euclidean } \mathbb{R}^2, \\ < 1 & \text{hyperbolic } \mathbb{H}^2. \end{cases} \quad (11)$$

There exists a triangle $T = T_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ in the respective space Ω_z ($\mathbb{S}^2$, $\mathbb{R}^2$ or $\mathbb{H}^2$) bounded by geodesics edges and with interior angles $\pi/\mathbf{p}$, $\pi/\mathbf{q}$, $\pi/\mathbf{r}$.

Theorem The group $G_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ is isomorphic to the group of isometries of Ω_z , generated by reflections on the 3 geodesic edges of $T \subset \Omega_z$. The triangle T is a fundamental domain for the action of $G_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ on Ω_z . $\square$

Let $H_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ be the index 2 subgroup of $G_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$, its elements are holomorphic automorphisms of Ω_z .

Corollary 4 1) *The action gives rise to a tessellation*

$$\mathcal{T}_{\mathbf{p}, \mathbf{q}, \mathbf{r}} = \{g(T) \mid g \in G_{\mathbf{p}, \mathbf{q}, \mathbf{r}}\}$$

of Ω_z , with blue tiles $\{g(T) \mid g \in H_{\mathbf{p}, \mathbf{q}, \mathbf{r}}\}$.

2) *There exist meromorphic functions $R : \Omega_z \longrightarrow \widehat{\mathbb{C}}$, each of degree the order of $H_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ and $q = 3$ critical values. Each function R and its tessellation $\mathcal{T}_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ are $H_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ -invariant.* $\square$

Example 3 (Schwarz's tessellations) As far as we known, the first examples of meromorphic functions that origin tessellations are due to H. A. Schwarz [29]. For each $\mathbf{p}, \mathbf{q}, \mathbf{r}$, as in Equation (11), there exists tessellation $\mathcal{T}_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ by triangles (tiles) $\{T_\alpha, T'_\alpha\}$ in $\mathbb{S}^2$, see Figure 5. Let $\mathfrak{P}$ be a k -dihedron or a platonic solid, see Figure 5 first column. We make the following conventions. The vertices $\{\mathbf{v}_\iota\} \in \mathfrak{P}$ (green points) are zeros of R ; the face centers $\mathbf{r} \in \mathfrak{P}$ (red points) are poles of R ; the middle points $\{\mathbf{e}_\kappa\}$ of edges of $\mathfrak{P}$ (yellow points) assume the value 1.

Moreover, we assign suitable multiplicities $\mu, \nu \geq 2$ to the vertices $\{\mathbf{v}_\iota\}$ and the centers $\{\mathbf{c}_j\}$, according to $\mathbf{p}, \mathbf{q}, \mathbf{r}$. The $H_{\mathbf{p}, \mathbf{q}, \mathbf{r}}$ -invariant functions are

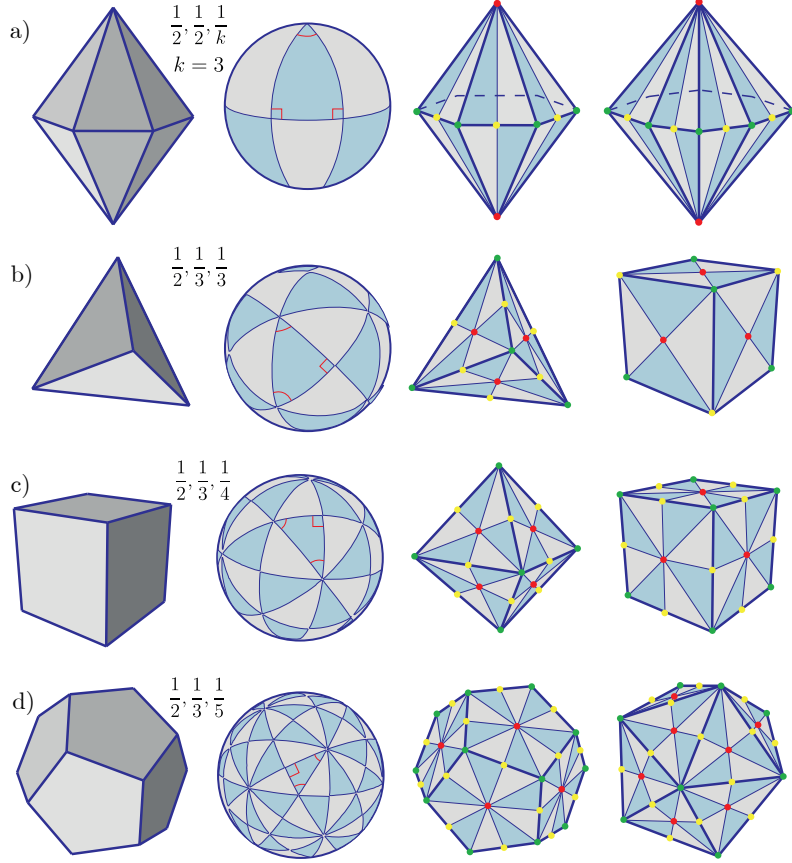


Fig. 5 Schwarz's tessellations. The left column illustrates the k -dihedron and the platonic solids. The other columns show tessellations of meromorphic G -invariant Belyĭ functions.

$$R(z) = \lambda \frac{\prod_{\iota} (z - \mathfrak{v}_{\iota})^{\mu_{\iota}}}{\prod_j (z - \mathfrak{c}_j)^{\mu_j}}, \quad \begin{cases} R(\mathfrak{v}_{\iota}) = 0, \\ R(\mathfrak{c}_j) = \infty, \\ R(\mathfrak{e}_{\kappa}) = 1. \end{cases}$$

They are Belyĭ functions with critical values $\{0, 1, \infty\}$. The explicit formulas for the G -invariant functions are cumbersome, for the icosahedron, we get $P(z)/Q(z)$ with

$$P(z) = z^5 \left(z^5 - (1 - \sqrt{5})^5 \right)^5 \left(z^5 - (1 + \sqrt{5})^5 \right)^5 \quad \text{and}$$

$$\begin{aligned}
Q(z) = & \left(z^5 - \left(\frac{1}{4} \left(-1 + \sqrt{5} + \sqrt{6(5 + \sqrt{5})} \right) + \frac{2i}{\sqrt{3} \left(1 - \frac{1}{\sqrt{15+6\sqrt{5}}} \right)} \right) \right)^5 \Bigg)^3 \\
& \left(z^5 - \left(-\frac{2\sqrt{3}(2+\sqrt{5})}{\sqrt{3}+3\sqrt{5}+2\sqrt{5}} - i \frac{2}{\sqrt{3} \left(1 + \frac{1}{\sqrt{15+6\sqrt{5}}} \right)} \right) \right)^5 \Bigg)^3 \\
& \left(z^5 - \left(-\frac{\sqrt{30(5+\sqrt{5})}}{15+\sqrt{75+30\sqrt{5}}} - i \frac{5\sqrt{18-6\sqrt{5}}}{15+\sqrt{75+30\sqrt{5}}} \right) \right)^5 \Bigg)^3 \\
& \left(z^5 - \left(\frac{1}{2} \left(2 + \sqrt{5} + \sqrt{15 + 6\sqrt{5}} \right) + i \sqrt{\frac{1}{2} \left(10 + \sqrt{5} + \sqrt{75 + 30\sqrt{5}} \right)} \right) \right)^5 \Bigg)^3.
\end{aligned}$$

Example 4 (Fortunate meromorphic functions) We say that a meromorphic function $R(z)$ is *fortunate* when its critical values belong to $\mathbb{R} \cup \{\infty\}$, in this case a natural choice for γ is this circle itself. Hence, a fortunate meromorphic function R has natural tessellation

$$M \setminus \Gamma, \text{ where } \Gamma = \{\Im(R(z)) = 0\}.$$

Some advantages of the tessellations of fortunate meromorphic functions are:

- The graph Γ is parametrized by the solution of the equations $\{R(z) = c \mid c \in \mathbb{R}\}$.
- For $M = \widehat{\mathbb{C}}$, as far as we know Γ is a reducible real algebraic curve of degree n (the degree of $R(z)$).

Let $\mathbb{Z}/p$ be the cyclic p -group, acting on $\mathbb{C}_z$ by rotations $\{r_s(z) \mid s = 1, \dots, p\}$. It is possible to construct fortunate functions with $(\mathbb{Z}/p)$ -symmetry, as follows. For $p \geq 2$, we use the family of meromorphic functions

$$R(z) = \int_0^z (P(\zeta)/Q(\zeta)) d\zeta$$

having the following properties:

- The critical points $\{z_\iota\}$ of $R(z)$ with their multiplicities are $\mathbb{Z}/p$ -invariant.
- The coefficients of $R(z)$ are real.
- For each critical point z_ι , there exists a rotation $r_s(z)$ such that $r_s(z_\iota) \in \mathbb{R}$ and its critical value can be expressed as a real integral, thus $\int_0^{z_\iota} (P(\zeta)/Q(\zeta)) d\zeta = \int_0^{r_s(z_\iota)} (P(\zeta)/Q(\zeta)) d\zeta$. Hence, such $R(z)$ is fortunate. In the other direction, a $\mathbb{Z}/p$ -invariant meromorphic function is not necessarily fortunate. Figure 6 illustrates the tessellations of the following fortunate functions

$$\begin{aligned}
a) \quad P(z) &= \int_0^z \left(\zeta^5(\zeta^3 - 1)^2(\zeta^3 + 1)^4(\zeta^6 - 2^6)^3 \right) d\zeta, & q = 5; \\
b) \quad R_1(z) &= \frac{z^5(z^5 - 4^5)^5(z^5 - 5^5)^5}{(z^5 - 1^5)^3(z^5 - 2^5)^2(z^5 - 3^5)}, & q = 7; \\
c) \quad R_2(z) &= \frac{(z^5 - 2^5)^5(z^5 - 4^5)^5}{z^5(z^5 + 1^5)^2(z^5 + 3^5)(z^5 + 5^5)^3}, & q = 7.
\end{aligned}$$

Example 5 1. Weierstrass $\wp$ -functions. Let $\Lambda = \{n + m\tau \mid (n, m) \in \mathbb{Z}^2, \Im(\tau) > 0\}$ be a lattice in $\mathbb{C}_z$. The associated Weierstrass $\wp$ -function on the Λ -torus M is

$$\wp(z) : M = \mathbb{C}_z/\Lambda \longrightarrow \widehat{\mathbb{C}}.$$

It has critical points $\mathcal{CP}_\wp = \{[0], [1/2], [\tau/2], [(1+\tau)/2]\}$, with critical values $\mathcal{CV}_\wp = \{\infty, e_1 = \wp(1/2), e_2, e_3\}$, where $\{e_k\}$ are the roots of the cubic equation $\{4e^3 - 60G_2e - 140G_3 = 0\}$, for

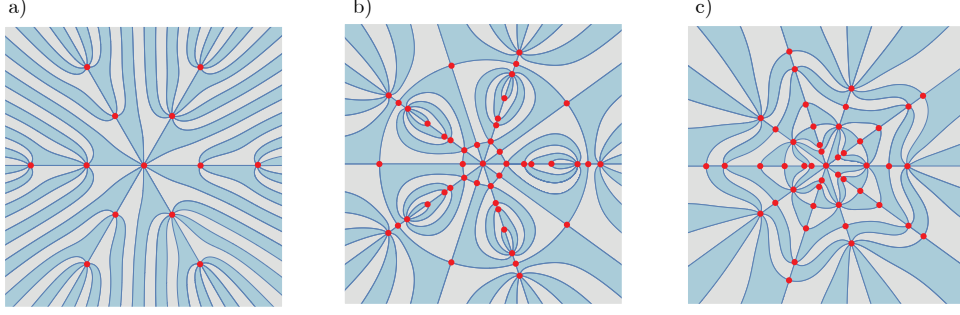


Fig. 6 Affine tessellations of three fortunate functions. a) A $\mathbb{Z}/3$ -invariant polynomial whose tiles are 5-gons. (b)–(c) Two $\mathbb{Z}/5$ -invariant meromorphic functions whose tiles are 7-gons (the cocritical points are not illustrated).

$G_k = \sum_{\omega} (1/\omega^{2k})$, $\omega \in \Lambda \setminus \{0\}$. The R-map $\hat{\Gamma}$ has $q = 4$, Figure 7.a describes the tessellations on a fundamental domain of the torus, and the consistent 4-labelling. The degree of $\wp(z)$ is 2. Clearly, the “chessboard” in Figure 7.a, can be generalized to a chessboard with $2\sigma_1$ rows and $2\sigma_2$ columns, for a pair $(\sigma_1, \sigma_2) \in \mathbb{N}^2$. Each pair gives rise to a new torus $\hat{M} = \mathbb{C}_z / \Lambda_{\sigma_1, \sigma_2}$, see Figure 7.b. The corresponding meromorphic function $\hat{R}$ on the torus $\hat{M}$ has degree $2\sigma_1\sigma_2$.

2. The tessellations of $\mathbb{C}_z$ by equilateral triangles, $p = q = r = 3$ in Equation (11), or more generally by two non isometric triangles T, T' , are obtained from the derivatives $\wp'(z)$, see Figure 7.c They originate from elliptic functions of degree 3 with $q = 3$ critical values on the corresponding torus $\mathbb{C}_z / \Lambda$.

3. Functions on hyperelliptic surfaces. Let $\hat{\Gamma} \subset \mathcal{M}$ be and R-map on a surface $\mathcal{M}$ of genus $g \geq 1$, the consistent 6-labelling is shown in Figure 7.d for genus 2. The tessellation $\mathcal{M} \setminus \hat{\Gamma}$ is by four $(2g+2)$ -gons with $2g+2$ vertices of valence 4. The associated meromorphic function R of degree 2 determines a hyperelliptic Riemann surface $M = (\mathcal{M}, J)$. There are $2g+2$ critical points of multiplicity 2 of R . For each genus $g \geq 1$, these meromorphic functions R attain the minimum number of critical points in $\mathcal{M}$. For a hyperelliptic Riemann surface

$$M = \{z^2 - (w - w_1) \cdots (w - w_{2g+2}) = 0\} \subset \mathbb{C}^2 = \{(w, z)\},$$

with $\{w_j \mid j \in 1, \dots, 2g+2\}$ distinct points, the projection $\pi_1 : M \rightarrow \mathbb{C}_w$ belongs to the above family of meromorphic functions.

4. Consider a “L-chessboard” as in Figure 7.e Under identification of the horizontal and vertical edges of the boundary, we get a surface M of genus 2 with a meromorphic function $R : M \rightarrow \hat{\mathbb{C}}$ of degree 6. There are 9 critical points of multiplicity 2 and 1 critical point of multiplicity 6. This example generalizes to any genus $g \geq 3$.

4.1 Closing comments

- In the case $\mathcal{M} = \mathbb{S}^2$, the existence of a consistent q -labelling for tessellations that arise from generic meromorphic functions, was proved in S. Koch *et al.* [18]; by proving that the tessellations with $2n$ tiles and $2n - 2$ vertices of valence 4 satisfy global and local balance conditions. The existence of a consistent q -labelling for tessellations given by generic polynomials is in L. J. González-Cely *et al.* [11].

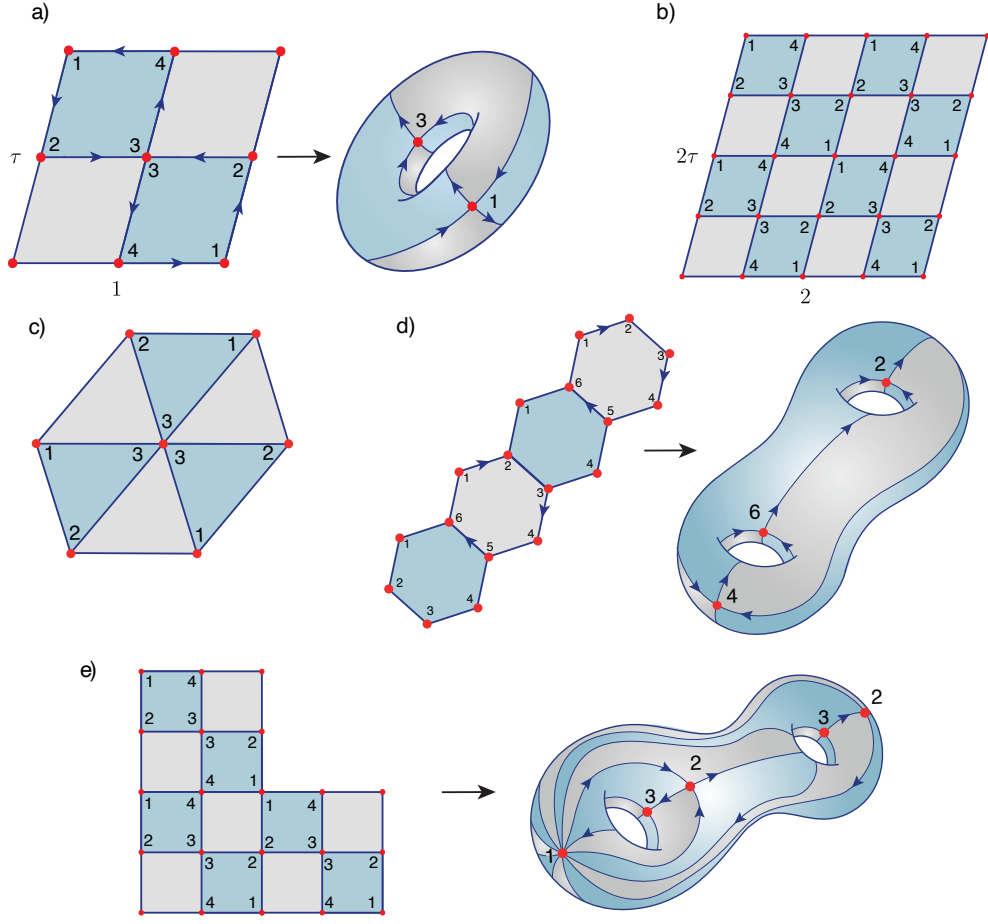


Fig. 7 Tessellations on compact Riemann surfaces and their consistent q -labellings. a) A Weierstrass function $\wp(z)$ on a torus. b) A lift of $\wp(z)$ to a covering torus. c) Derivative $\wp'(z)$ on a torus. d) A meromorphic function on a hyperelliptic surface. e) A meromorphic function on a surface of genus 2.

For the non-generic case in 2015, J. Tomasini [33] proved a characterization for general rational functions. He follows an equivalent approach by considering an intermediate between tessellations or their duals the Speiser graphs. Compare also with the unpublished work of A. Lobato, that studies a generalization of [18] to arbitrary genus and finite branched coverings of the sphere, see [22], [23].

Finally, A. Alvarez-Parrilla *et al.* in [2] provide a characterization for Speiser functions: these include infinitely branched coverings of the sphere. Their work is based on Speiser graphs (the dual of the Tessellations) and the *Bipartite Transportation Problem*, well known in graph theory.

- Given a tessellation $\mathcal{T}$ of $\mathcal{M}$ as in assertion (2) of Theorem 1; how can we provide an analytic description (in terms of coefficients for instance) of a corresponding R on M ?
- Consider a non homogeneous tessellation $\mathcal{T}$ on $\mathcal{M}$ that admits a consistent q -labelling $\mathcal{L}_c$, thus $\mathcal{T}$ is realizable from a meromorphic function. Several natural questions arise:

How many different consistent q -labellings are there?

Under what conditions the value of q is unique?

Example 2 already shows these phenomena.

- Consider the space of analytic objects $\{(M, R, \mathcal{L}_\gamma)\}$, where the genus of M , the degree of R and the number of critical values q are fixed. Considering the configurations of critical values $\{w_1, \dots, w_q\}$ of R as free parameters, and recalling that $\mathcal{L}_\gamma$ is the cyclic order in the critical values of R , it is clear that $\mathcal{L}_\gamma$ is a discrete quantity. Very roughly speaking, using Diagram 10, we have the map

$$\textcircled{3} \circ \textcircled{5} : \{(M, R, \mathcal{L}_\gamma)\} \longrightarrow \{(\mathcal{M}, \Gamma, \mathcal{L}_c, \{w_1, \dots, w_q\})\}.$$

Thus we have a certain kind of fibration, with different topological fibers; from a family of analytic objects to topological data. The map $\textcircled{6}$ is a section of this fibration. An accurate formulation of these maps and their study is a future project.

- A meromorphic function $w : \mathbb{C}_z \longrightarrow \widehat{\mathbb{C}}$ is Speiser when it has a finite number of critical and asymptotic values, *i.e.* it belongs to the family $F(w_1, \dots, w_q)$. The study of Speiser functions $w(z)$ on simply connected Riemann surfaces was considered by A. Speiser [30] and R. Nevanlinna [27, Ch. XI]. The study of tessellations and applications to differential equations is the subject of A. Alvarez-Parrilla *et al.* [3].

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